

Computer Science & IT

COMPILER DESIGN (CD)

Live Variable Analysis

One Shot

By- Ravindra Sir



Recap of Previous Lecture



Topic

Code Optimization

Topic

Topic

Topics to be Covered



Topic

LVA

Topic

Topic

Join Telegram





Inspiring Stories : Mula Gabharu

Mula Gabharu – The Queen Who Raised Women Soldiers

Background: Mula Gabharu was the daughter of an Ahom king in Assam in the early 1500s.

Struggles: Her husband fell fighting an invading army. Her people were helpless and losing hope.

Achievements: She trained women warriors, led them into battle with courage, and confronted the enemy herself before being martyred.

Impact: If she could turn grief into action and lead women into battle, you can turn doubt into study and lead your rank up.



Inspiring Stories : B. N. Jagatap

B. N. Jagatap

Background: Grew up in a modest family in rural Maharashtra with no strong science background at home.

Education: PhD in Nuclear Magnetic Resonance (NMR) from IISc Bangalore.

Career Achievements: Improved NMR methods to study molecules in great detail, helping drug design and materials research.

Impact: Made high-end science tools more effective for Indian research labs and universities.



Inspiring Stories : Veeramangai Kuyili

Veeramangai Kuyili – The Woman Who Became a Human Torch

Background: Kuyili was a trusted soldier of queen Velu Nachiyar from Tamil Nadu in the 1700s.

Struggles: She worked without praise and lived in a time when women were rarely seen in battle.

Achievements: During one attack, she covered her body in ghee, set herself on fire, and jumped into the British armory. Her act became a key turning point in the war.

Impact: If she could light herself on fire to save her queen, you can light up that answer sheet with focus



Inspiring Stories : K. J. Rao

K. J. Rao

Background: Born in a small town in Karnataka with very limited means. Had to work hard to afford studies.

Education: PhD in Solid State Chemistry from IISc Bangalore.

Career Achievements: Developed new ceramic materials that can store energy and work in extreme heat.

Impact: Helped India make better energy storage systems and heat-resistant tools for industry.

The main purpose of liveness analysis is

- 1) Register allocation
- 2) Dead code elimination

1

2 X = Y + 1 ✓

3

4

5

R_0	R_1	R_2	R_3
a	b	c	d y

$a, b, c \rightarrow$ live at line 2

$d \rightarrow$ dead

1 $\boxed{a = (b + c)}$ ✓ $\rightarrow b, c$

2 $\boxed{x = y + z}$ ✓

3 $\boxed{m = n + k}$ ✓

4 $\boxed{c = (p) + 1}$ ✓

✓ $a = b + c$
 $\underbrace{\quad}$ $\underbrace{\quad}$
 define use

Definition of a live variable:-

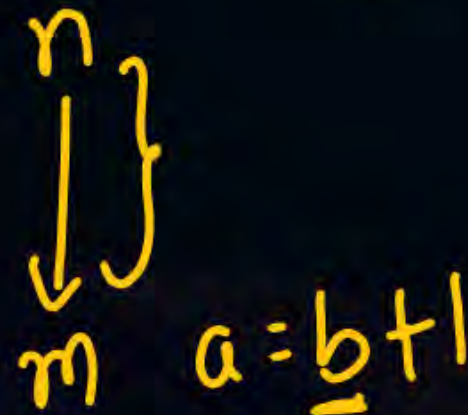
→ A variable is said to be live at a statement, if it is used in the statement

$$(a) = (b + c)$$

(A)

→ we say that a variable 'v' is live at a statement 'n', if there exists a path in the control flow graph from line 'n' to a statement 'm' such that 'v' is used in 'm' and for each $n \leq k < m$

$$v \notin \text{def}[k]$$



- 1) $\underline{v} = 1$
- 2) $\underline{z} = \underline{v} + 1$
- 3) $\underline{x} = \underline{z} * \underline{v}$
- 4) $\underline{y} = \underline{x} + 2$
- 5) $\underline{w} = \underline{x} + \underline{y} + \underline{z}$
- 6) $\underline{u} = \underline{z} + 2$
- 7) $\underline{v} = \underline{u} + \underline{w} + \underline{y}$
- 8) return $v * u$

	$\checkmark$ x	y	z	u	v	w
1	D	D	D	D	D	D
2	D	D	D	D	L	D
$\checkmark$ 3	D	D	L	D	L	D
4	L	D	L	D	D	D
5	L	L	L	D	D	D
6	D	L	L	D	D	L
7	D	L	D	L	D	L
8	D	D	D	L	L	D

THANK - YOU